

ZACHARY TIANYI TANG

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RESEARCH INTERESTS

I believe the path to genuinely useful robots depends on highly sample-efficient post-training: methods that allow foundational models to adapt to new human environments with minimal demonstration (<1 hour), and the ability to reason about the consequences of one's actions. I am currently interested in using signal-rich **HIL / intervention data** for **sample-efficient post-training**, and building scalable **4D world models** for **physical counterfactual search/reasoning**.

EDUCATION

University of Toronto

B.S., Computer Science; GPA: 3.77/4.00

Toronto, ON

Expected May 2028

RESEARCH EXPERIENCE

RLLAB, Yonsei University (Prof. Youngwoon Lee)

Research Intern

Seoul, South Korea

May 2026 - Present

- Driving lead-author research with my advisor on **sample-efficient post-training** for pretrained robot policies, studying **HIL/DAGger intervention quality**, timing, correction trajectories, and OOD/uncertainty signals for robot-gating
- Helped Yonsei achieve **top-8 finishes** in both tracks of the **RSS 2026 Post-Training for Robotics Foundation Models Challenge** via **RL-Token-style offline post-training** on failure/intervention rollouts

Dynamic Optimization & RL Lab, University of Toronto (Prof. Chi-Guhn Lee)

Researcher

Toronto, ON

Sep 2025 - Apr 2026

- Co-authored **WireCraft** (Zhu et al., 2026, arXiv preprint), a simulation benchmark for **industrial DLO manipulation** spanning connector insertion, clip routing, and channel seating, with configurable fixtures and assets
- Engineered **IsaacLab** infrastructure for articulated and FEM-based deformable wire simulation, UR5 manipulation tasks, vectorized RL training, and scalable trajectory generation
- Trained privileged-state **RL policies** for wire/clip routing, developing staged rewards, success metrics, and scripted grasping policies while debugging contact-rich **PhysX instability** across thousands of parallel environments

Human Computer Lab (backed by South Park Commons, FR8)

Software Engineer (R&D)

Remote

Jan 2026 - Mar 2026

- Developed **robot-embodied game prototypes** with Raspberry Pi, WebSockets, and browser-based controls, translating physical lamp/robot movement into aiming, camera, reload, and lighting feedback mechanics
- Explored **human-robot interaction for AI worlds** by connecting a physical lamp interface to Claw Town, a multi-agent sandbox with autonomous NPC actions, chat, emotes, heartbeat loops, and goal-directed behavior

Intelligent Robot Learning Lab, University of Alberta (Prof. Matthew Taylor)

Research Lead

Remote

Sep 2025 - Present

- Initiated project, recruited full undergraduate research team, and secured supervision from **Prof. Matthew Taylor** for a novel challenging Overcooked benchmark on **partially observable multi-agent RL communication**
- Received **People's Choice Award** at **CUCAI 2026**, Canada's largest undergraduate AI conference; preparing submission to **AAAI 2027**

PROJECTS & EXTRACURRICULARS

Heimer Academy | [React](#), [Vite](#), [FastAPI](#), [Supabase](#), [Amazon SageMaker](#), [Amazon Bedrock](#) ([Devpost](#)) ([GitHub](#))

- 1st Place (\$10,000 USD + \$10k AWS Credits)** at **AWS x Riot Games Rift Rewind** global hackathon
- Engineered champion recommendation system using **co-occurrence graph analysis** across tens of thousands of player profiles and **pairwise ability similarity scoring** over **600+ abilities (200,000+ comparisons)**

Identity Matrix | [React](#), [TypeScript](#), [Phaser](#), [FastAPI](#), [Supabase](#), [WebSockets](#) ([Devpost](#)) ([GitHub](#))

- Overall 1st Place (\$2,000 USD prize)** and **Best "UofT" Hack** at **UofTHacks 13**
- Built Simulacra-inspired real-time multiplayer world where offline users become persistent LLMs agents with personality and memory

TECHNICAL SKILLS

Programming: Python, C++ , JavaScript, TypeScript

AI/Robotics: PyTorch, JAX, LeRobot, Hugging Face, OpenCV, NumPy, SciPy, IsaacSim, IsaacLab

Systems & Tools: CUDA, TensorRT, SLURM, Docker, Git, Linux, Bash

Jun 2026